CS3103 | From Request to Response

Student quick reference • HTTP / JSON / Spring Boot

Learning objectives

Trace a client–server exchange; recognize a GET and POST handler; predict and test 200, 201, 400, and 404 responses.

1. HTTP in one glance

REQUEST	MEANING	TYPICAL RESPONSE
GET /api/trips/101	Read existing trip	200 OK + JSON object
GET /api/trips/999	Read missing trip	404 Not Found
POST /api/trips	Create a new trip	201 Created + Location
POST blank origin	Invalid submitted data	400 Bad Request

2. Request anatomy

Method + path + headers + optional body. Content-Type describes the representation being sent; Accept describes preferred response formats.

3. Response anatomy

HTTP status + response headers + optional JSON body. A 404 is an HTTP response; it is not automatically a network outage.

4. Example JavaScript

```
const r = await fetch("/api/trips/101");
if (!r.ok) { /* handle HTTP status */ }
else { const trip = await r.json(); /* render */ }
```

Activity: Predict before you run

A. The client requests a nonexistent trip. What method, path, and status do you expect?

B. The client creates a trip from Moncton to Fredericton. What request method, headers, and body should it send?

C. A POST body has an empty origin. What should the API report?

Mini lab (after class)

Implement GET /api/trips in the provided starter. Verify valid and missing reads, creation, and invalid input with curl, Postman, or the browser UI. Submit a short test record showing method, URL, status, and expected response.